

Song Shi

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EDUCATION

The University of British Columbia
Bachelor of Science in Computer Science

Vancouver, BC
November 2025

EXPERIENCE

Undergraduate Research Assistant

June 2023 – August 2023

UBC Statistics Department

Vancouver, BC

- Conducted research related to the relative bias and variance of nearest neighbour imputation(NNI) for survey data under the faculty supervision.
- Coded in R to perform NNI and different variants of the imputation method to compare against NNI.
- Implemented multiple NNI variants in R and evaluated performance across one-dimensional and multi-dimensional predictor settings.
- Explored the relative bias of the multi-dimensional predictor case of NNI.
- Authored a technical report detailing methodology, experimental results, and statistical interpretation.

PROJECTS

RAG Text Generation System | *FastAPI, Next.js, Qdrant, Docker, PostgreSQL*

March 2026 – present

- Designed and implemented a custom retrieval-augmented generation (RAG) pipeline to transform input text into a target philosophical writing style (Hegelian)
- Integrated Qdrant vector database for embedding-based semantic search over domain-specific corpus
- Developed FastAPI backend to orchestrate retrieval, prompt construction, and LLM-based text generation
- Containerized full-stack system using Docker Compose, debugging cross-container issues
- Explored sandboxing of AI agents by isolating CLI execution within containers for safer automated workflows

N-Body Simulation | *Next.js, Three.js, GLSL, GPGPU*

February 2026 – February 2026

- Built a real-time N-body gravitational simulation using WebGL and Three.js
- Simulated 15,000 particles at real-time frame rates
- Used ping-pong render targets for position/velocity state updates and explicit Euler integration
- Offloaded pairwise force calculations to GPU using GPGPU (compute shader parallelization)
- Achieved 60 FPS on consumer laptop GPU

Clickbait prediction research | *Python, Pytorch, LaTeX*

April 2025 – April 2025

- Machine Learning Research done in a team of 2, predicting Clickbait articles, using the title as a predictor
- Created custom 128-dimensional word embeddings to represent textual data for optimized sequence learning
- Built a PyTorch-based RNN model on GPU to classify clickbait articles using titles as input, improving prediction accuracy through iterative tuning.
- Conducted comparative analysis of multiple machine learning models and documented performance results in a technical report
- Engineered advanced data preprocessing pipelines and incorporated regularization techniques to enhance model robustness and mitigate overfitting for clickbait title prediction tasks.

OCR System | *FastAPI, Python, React, Next.js, PostgreSQL, Docker, Git*

January 2025 – April 2025

- Contribute in a team of 9 for an OCR document upload WEB app.
- Utilize AWS S3 and Lambda for a serverless app
- Implement a SSR frontend using NextJS.
- Utilize Docker containerization for local testing and deployment.
- Use a CI pipeline to automate testing and build

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, TypeScript, HTML/CSS, R, GLSL

Frameworks: React, Node.js, Next.js, JUnit, FastAPI, OpenGL, WebGL

Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Libraries: pandas, NumPy, Pytorch, Three.js, SQLAlchemy, Pydantic, LangChain